Atiyah–Macdonald, Chapter 4: Primary Decomposition

Solutions to the exercises

Conventions. For an ideal $\mathfrak{a}$ and multiplicatively closed S, $S(\mathfrak{a}) = (S^{-1}\mathfrak{a})^c = \bigcup_{s\in S}(\mathfrak{a} : s)$ is the contraction of $S^{-1}\mathfrak{a}$ (3.11(ii)), and $S_{\mathfrak{p}}(\mathfrak{a})$ means $S(\mathfrak{a})$ for $S = A - \mathfrak{p}$. A *decomposable* ideal is one with a primary decomposition; *minimal* (irredundant) decompositions have distinct radicals $\mathfrak{p}_i = r(\mathfrak{q}_i)$ and no superfluous component. From the text I use: $r(\mathfrak{q})$ is prime for $\mathfrak{q}$ primary (4.1); powers of a maximal ideal are primary (4.2); 4.3 and 4.4; the first uniqueness theorem 4.5 (the prime ideals $r(\mathfrak{a} : x)$ that occur are exactly the $\mathfrak{p}_i$) and 4.6; 4.7, 4.8, 4.9 and the second uniqueness theorem 4.10–4.11; also 1.11 (prime avoidance), 1.13, 1.14 and Ch. 3.

This chapter has 23 end-of-chapter exercises and no numbered in-text ones.

End-of-chapter exercises

Exercise 1

> **Problem.** If an ideal $\mathfrak{a}$ has a primary decomposition, then $\operatorname{Spec}(A/\mathfrak{a})$ has only finitely many irreducible components.

Solution. Let $\mathfrak{a} = \bigcap_{i=1}^{n} \mathfrak{q}_i$ be a minimal primary decomposition, $\mathfrak{p}_i = r(\mathfrak{q}_i)$. $\operatorname{Spec}(A/\mathfrak{a}) \cong V(\mathfrak{a})$ (Ch. 1, Ex. 21(iv)) and the irreducible components of $\operatorname{Spec}(A/\mathfrak{a})$ are the closed sets $V(\bar{\mathfrak{p}})$ with $\bar{\mathfrak{p}}$ a minimal prime of $A/\mathfrak{a}$ (Ch. 1, Ex. 20(iv)), i.e. $V(\mathfrak{p})$ with $\mathfrak{p}$ minimal among the primes containing $\mathfrak{a}$. By 4.6 these are among $\mathfrak{p}_1, \ldots, \mathfrak{p}_n$. So there are at most n components. □

Exercise 2

> **Problem.** If $\mathfrak{a} = r(\mathfrak{a})$, then $\mathfrak{a}$ has no embedded prime ideals.

Solution. (We assume $\mathfrak{a}$ is decomposable, so that associated primes make sense.) Let $\mathfrak{a} = \bigcap \mathfrak{q}_i$ be a minimal decomposition, $\mathfrak{p}_i = r(\mathfrak{q}_i)$, so $r(\mathfrak{a}) = \bigcap \mathfrak{p}_i$. Every embedded $\mathfrak{p}_j$ contains a minimal $\mathfrak{p}_i$ (4.6 and the definition), so we may drop the embedded ones from the intersection: $\mathfrak{a} = r(\mathfrak{a}) = \bigcap_{\mathfrak{p}_i \text{ minimal}} \mathfrak{p}_i$. This is a primary decomposition by prime ideals (primes are primary), with pairwise incomparable (so irredundant) distinct radicals, i.e. a minimal primary decomposition. By the first uniqueness theorem 4.5, the associated primes of $\mathfrak{a}$ are the $\mathfrak{p}_i$ appearing in *any* minimal decomposition; hence they are all minimal, and there are no embedded primes. □

Exercise 3

Problem. If A is absolutely flat, every primary ideal is maximal.

Solution. Let $\mathfrak{q}$ be primary and $\mathfrak{p} = r(\mathfrak{q})$. $A/\mathfrak{N}$ is absolutely flat (a homomorphic image, Ch. 2, Ex. 28), so every prime of A is maximal (Ch. 3, Ex. 11); in particular $\mathfrak{p}$ is maximal. Let $x \in \mathfrak{p}$. As A is absolutely flat, $x = ax^2$ for some a (Ch. 2, Ex. 27), so $x(1-ax) = 0 \in \mathfrak{q}$. Now $1-ax \notin \mathfrak{p}$ (else $1 \in \mathfrak{p}$), hence $(1-ax)^n \notin \mathfrak{q}$ for all n; since $\mathfrak{q}$ is primary, $x \in \mathfrak{q}$. Thus $\mathfrak{p} \subseteq \mathfrak{q} \subseteq \mathfrak{p}$ and $\mathfrak{q} = \mathfrak{p}$ is maximal. □

Exercise 4

Problem. In the polynomial ring $\mathbf{Z}[t]$, the ideal $\mathfrak{m} = (2, t)$ is maximal and the ideal $\mathfrak{q} = (4, t)$ is $\mathfrak{m}$-primary, but is not a power of $\mathfrak{m}$.

Solution. $\mathbf{Z}[t]/(2,t) \cong \mathbf{Z}/2 = \mathbf{F}_2$, so $\mathfrak{m}$ is maximal. We have $\mathfrak{m}^2 = (4, 2t, t^2) \subseteq \mathfrak{q} = (4, t) \subseteq \mathfrak{m}$, hence $r(\mathfrak{q}) = \mathfrak{m}$ and $\mathfrak{q}$ is $\mathfrak{m}$-primary by 4.2.

$\mathfrak{q}$ is not a power of $\mathfrak{m}$: $\mathfrak{q} \neq \mathfrak{m}$ since $2 \notin \mathfrak{q}$ (note $\mathfrak{q} \cap \mathbf{Z} = 4\mathbf{Z}$); $\mathfrak{q} \neq \mathfrak{m}^2$ since $t \in \mathfrak{q}$ but $t \notin \mathfrak{m}^2$ (the coefficient of t in an element $4f + 2tg + t^2h$ of $\mathfrak{m}^2$ is even). Since $\mathfrak{m}^n \subseteq \mathfrak{m}^2 \subsetneq \mathfrak{q}$ for $n \geq 2$, $\mathfrak{q}$ equals no $\mathfrak{m}^n$. So $\mathfrak{m}^2 \subsetneq \mathfrak{q} \subsetneq \mathfrak{m}$. □

Exercise 5

Problem. In the polynomial ring $K[x, y, z]$ where K is a field and x, y, z are independent indeterminates, let $\mathfrak{p}_1 = (x, y)$, $\mathfrak{p}_2 = (x, z)$, $\mathfrak{m} = (x, y, z)$; $\mathfrak{p}_1$ and $\mathfrak{p}_2$ are prime, and $\mathfrak{m}$ is maximal. Let $\mathfrak{a} = \mathfrak{p}_1\mathfrak{p}_2$. Show that $\mathfrak{a} = \mathfrak{p}_1 \cap \mathfrak{p}_2 \cap \mathfrak{m}^2$ is a reduced primary decomposition of $\mathfrak{a}$. Which components are isolated and which are embedded?

Solution. $\mathfrak{a} = \mathfrak{p}_1\mathfrak{p}_2 = (x^2, xz, xy, yz)$. We have $\mathfrak{p}_1 \cap \mathfrak{p}_2 = (x, yz)$ (monomial ideals: a monomial lies in both iff it is divisible by x or by both y and z), and $\mathfrak{m}^2 = (x^2, y^2, z^2, xy, xz, yz)$. Intersecting monomial ideals: $(x, yz) \cap \mathfrak{m}^2 = (x) \cap \mathfrak{m}^2 + (yz) \cap \mathfrak{m}^2 = (x^2, xy, xz) + (yz) = \mathfrak{a}$. So $\mathfrak{a} = \mathfrak{p}_1 \cap \mathfrak{p}_2 \cap \mathfrak{m}^2$.

Primary: $\mathfrak{p}_1, \mathfrak{p}_2$ are prime, $\mathfrak{m}^2$ is $\mathfrak{m}$-primary (4.2). The radicals $\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{m}$ are distinct.

Irredundant: $z^2 \in \mathfrak{p}_2 \cap \mathfrak{m}^2$ but $z^2 \notin \mathfrak{p}_1$, so $\mathfrak{p}_2 \cap \mathfrak{m}^2 \not\subseteq \mathfrak{p}_1$; $y^2 \in \mathfrak{p}_1 \cap \mathfrak{m}^2$, $y^2 \notin \mathfrak{p}_2$; $x \in \mathfrak{p}_1 \cap \mathfrak{p}_2$, $x \notin \mathfrak{m}^2$. So no component can be omitted: the decomposition is reduced (minimal).

Isolated vs. embedded: $\mathfrak{p}_1, \mathfrak{p}_2$ are incomparable and minimal; $\mathfrak{m} \supsetneq \mathfrak{p}_1$. So $\mathfrak{p}_1, \mathfrak{p}_2$ are the isolated primes, and the component $\mathfrak{m}^2$ (belonging to $\mathfrak{m} = (x, y, z)$) is embedded. □

Exercise 6

Problem. Let X be an infinite compact Hausdorff space, $C(X)$ the ring of real-valued continuous functions on X (Chapter 1, Exercise 26). Is the zero ideal decomposable in this ring?

Solution. *No.*

Step 1: X contains infinitely many pairwise disjoint non-empty open sets $U_1, U_2, \ldots$. An infinite subset $E \subseteq X$ has an accumulation point x_0 (compactness). Choose $y_1 \in E - \{x_0\}$ and disjoint open $U_1 \ni y_1$, $V_1 \ni x_0$. V_1 contains infinitely many points of E, so choose $y_2 \in E \cap V_1 - \{x_0\}$ and disjoint open $U_2 \ni y_2$, $V_2 \ni x_0$ with $U_2, V_2 \subseteq V_1$; and so on. Then $U_n \subseteq V_{n-1} \subseteq V_k$ for $k < n$, and $U_k \cap V_k = \emptyset$, so the U_n are pairwise disjoint.

Step 2: functions. X is normal, so by Urysohn there are continuous $f_n : X \to [0,1]$ with $f_n(y_n) = 1$ and $f_n = 0$ outside U_n. Then $f_n \neq 0$ and $f_n f_m = 0$ for $n \neq m$.

Step 3. Suppose $0 = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_r$ with $\mathfrak{q}_i$ primary, $\mathfrak{p}_i = r(\mathfrak{q}_i)$ prime. Then $\bigcap \mathfrak{p}_i = r(0) = 0$ ($C(X)$ has no non-zero nilpotents). Fix i. If $n \neq m$ and neither f_n nor f_m lies in $\mathfrak{p}_i$, then $f_n f_m = 0 \in \mathfrak{p}_i$ contradicts primality. So at most one n has $f_n \notin \mathfrak{p}_i$; i.e. $\{n : f_n \in \mathfrak{p}_i\}$ is cofinite. A finite intersection of cofinite sets is cofinite, hence non-empty: some f_n lies in every $\mathfrak{p}_i$, so $f_n \in \bigcap \mathfrak{p}_i = 0$, contradicting $f_n \neq 0$.

Hence 0 is not decomposable in $C(X)$. □

Exercise 7

> **Problem.** Let A be a ring and let $A[x]$ denote the ring of polynomials in one indeterminate over A. For each ideal $\mathfrak{a}$ of A, let $\mathfrak{a}[x]$ denote the set of all polynomials in $A[x]$ with coefficients in $\mathfrak{a}$.
>
> (i) $\mathfrak{a}[x]$ is the extension of $\mathfrak{a}$ to $A[x]$.
>
> (ii) If $\mathfrak{p}$ is a prime ideal in A, then $\mathfrak{p}[x]$ is a prime ideal in $A[x]$.
>
> (iii) If $\mathfrak{q}$ is a $\mathfrak{p}$-primary ideal in A, then $\mathfrak{q}[x]$ is a $\mathfrak{p}[x]$-primary ideal in $A[x]$. [Use Chapter 1, Exercise 2.]
>
> (iv) If $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$ is a minimal primary decomposition in A, then $\mathfrak{a}[x] = \bigcap_{i=1}^n \mathfrak{q}_i[x]$ is a minimal primary decomposition in $A[x]$.
>
> (v) If $\mathfrak{p}$ is a minimal prime ideal of $\mathfrak{a}$, then $\mathfrak{p}[x]$ is a minimal prime ideal of $\mathfrak{a}[x]$.

Solution. (i) The extension is the ideal $A[x]\mathfrak{a}$; its elements $\sum g_i a_i$ have all coefficients in $\mathfrak{a}$, and conversely $\sum a_j x^j = \sum x^j a_j \in A[x]\mathfrak{a}$.

(ii) $\mathfrak{p}[x] = \mathrm{Ker}(A[x] \to (A/\mathfrak{p})[x])$ and $(A/\mathfrak{p})[x]$ is a domain.

(iii) $A[x]/\mathfrak{q}[x] \cong (A/\mathfrak{q})[x]$ and $\mathfrak{q}[x] \neq A[x]$. By Ch. 1, Ex. 2(iii), a zero-divisor $f = \sum \bar{a}_j x^j$ of $(A/\mathfrak{q})[x]$ is killed by some $\bar{a} \neq 0$ in $A/\mathfrak{q}$; then $\bar{a}\bar{a}_j = 0$ for all j, so each $\bar{a}_j$ is a zero-divisor of $A/\mathfrak{q}$ (or 0), hence nilpotent, since $\mathfrak{q}$ is primary; by Ch. 1, Ex. 2(ii), f is nilpotent. So $\mathfrak{q}[x]$ is primary. Its radical: $f \in r(\mathfrak{q}[x]) \iff \bar{f}$ nilpotent in $(A/\mathfrak{q})[x] \iff$ all $\bar{a}_j$ nilpotent (Ex. 2(ii)) $\iff a_j \in r(\mathfrak{q}) = \mathfrak{p}$, i.e. $r(\mathfrak{q}[x]) = \mathfrak{p}[x]$.

(iv) A polynomial has all coefficients in $\bigcap \mathfrak{q}_i$ iff it has all coefficients in each $\mathfrak{q}_i$, so $\mathfrak{a}[x] = \bigcap \mathfrak{q}_i[x]$; the $\mathfrak{q}_i[x]$ are $\mathfrak{p}_i[x]$-primary by (iii) and $\mathfrak{p}_i \neq \mathfrak{p}_j \Rightarrow \mathfrak{p}_i[x] \neq \mathfrak{p}_j[x]$ (compare constants). Irredundance: if $\big(\bigcap_{j\neq i} \mathfrak{q}_j\big)[x] \subseteq \mathfrak{q}_i[x]$, comparing constant polynomials gives $\bigcap_{j\neq i} \mathfrak{q}_j \subseteq \mathfrak{q}_i$, a contradiction.

(v) The minimal primes of $\mathfrak{a}$ are the minimal elements of $\{\mathfrak{p}_1, \ldots, \mathfrak{p}_n\}$ (4.6), and the minimal primes of $\mathfrak{a}[x]$ are the minimal elements of $\{\mathfrak{p}_1[x], \ldots, \mathfrak{p}_n[x]\}$ by (iv) and 4.6. Since $\mathfrak{p}_i \subseteq \mathfrak{p}_j \iff \mathfrak{p}_i[x] \subseteq \mathfrak{p}_j[x]$, a minimal $\mathfrak{p}_i$ gives a minimal $\mathfrak{p}_i[x]$. □

Exercise 8

Problem. Let k be a field. Show that in the polynomial ring $k[x_1, \ldots, x_n]$ the ideals $\mathfrak{p}_i = (x_1, \ldots, x_i)$ $(1 \leq i \leq n)$ are prime and all their powers are primary. [Use Exercise 7.]

Solution. $k[x_1, \ldots, x_n]/\mathfrak{p}_i \cong k[x_{i+1}, \ldots, x_n]$ is a domain, so $\mathfrak{p}_i$ is prime. We show by induction on n that every power $\mathfrak{p}_i^m$ is $\mathfrak{p}_i$-primary in $A_n = k[x_1, \ldots, x_n]$. For $i = n$, $\mathfrak{p}_n$ is maximal and its powers are primary (4.2); this covers $n = 1$. Let $i < n$ and $A_n = A_{n-1}[x_n]$. The extension of $\mathfrak{p}_i^m \subseteq A_{n-1}$ to A_n is $(\mathfrak{p}_i^m)[x_n]$ (Ex. 7(i)), and it equals the m-th power of the extension $\mathfrak{p}_i[x_n] = (x_1, \ldots, x_i)A_n$. By induction $\mathfrak{p}_i^m$ is $\mathfrak{p}_i$-primary in A_{n-1}, so by Ex. 7(iii) $(\mathfrak{p}_i^m)[x_n]$ is $\mathfrak{p}_i[x_n]$-primary in A_n, i.e. the m-th power of $\mathfrak{p}_i$ in A_n is $\mathfrak{p}_i$-primary. □

Exercise 9

Problem. In a ring A, let $D(A)$ denote the set of prime ideals $\mathfrak{p}$ which satisfy the following condition: there exists $a \in A$ such that $\mathfrak{p}$ is minimal in the set of prime ideals containing $(0 : a)$. Show that $x \in A$ is a zero-divisor $\iff$ $x \in \mathfrak{p}$ for some $\mathfrak{p} \in D(A)$.

Let S be a multiplicatively closed subset of A, and identify $\mathrm{Spec}(S^{-1}A)$ with its image in $\mathrm{Spec}(A)$ (Chapter 3, Exercise 21). Show that $D(S^{-1}A) = D(A) \cap \mathrm{Spec}(S^{-1}A)$.

If the zero ideal has a primary decomposition, show that $D(A)$ is the set of associated prime ideals of 0.

Solution. *Zero-divisors.* ($\Rightarrow$) If $xa = 0$, $a \neq 0$, then $x \in (0 : a) \neq (1)$. $r(0 : a)$ is the intersection of the primes containing $(0 : a)$, hence of the minimal ones among them; so x lies in each such minimal prime, which belongs to $D(A)$ (and at least one exists).

($\Leftarrow$) Let $\mathfrak{p}$ be minimal over $\mathfrak{b} = (0 : a)$ and $x \in \mathfrak{p}$. The only prime of $A_\mathfrak{p}$ containing $\mathfrak{b}_\mathfrak{p}$ is $\mathfrak{p}A_\mathfrak{p}$, so $\mathfrak{p}A_\mathfrak{p} = r(\mathfrak{b}_\mathfrak{p})$, and $(x/1)^n \in \mathfrak{b}_\mathfrak{p}$ for some n: there is $s \notin \mathfrak{p}$ with $sx^n \in (0 : a)$, i.e. $b := sa$ satisfies $x^n b = 0$. Moreover $b \neq 0$: otherwise $s \in (0 : a) \subseteq \mathfrak{p}$. Let $k \geq 1$ be minimal with $x^k b = 0$; then $x^{k-1} b \neq 0$ and $x \cdot x^{k-1} b = 0$. So x is a zero-divisor.

Localization. Let $\mathfrak{q} \in \mathrm{Spec}(S^{-1}A)$, $\mathfrak{p} = \mathfrak{q}^c$ $(\mathfrak{p} \cap S = \emptyset)$. For $a \in A$, $s \in S$ we have $\mathrm{Ann}_{S^{-1}A}(a/s) = \mathrm{Ann}_{S^{-1}A}(a/1) = S^{-1}(0 : a)$ (3.14 applied to the cyclic module Aa). Primes of $S^{-1}A$ containing $S^{-1}(0 : a)$ correspond (order-preservingly, 3.11) to primes of A disjoint from S containing $(0 : a)$, and any prime contained in $\mathfrak{p}$ is disjoint from S. Hence $\mathfrak{q}$ is minimal over $S^{-1}(0 : a)$ iff $\mathfrak{p}$ is minimal over $(0 : a)$ among all primes of A (for "$\Leftarrow$": take $\mathfrak{p}$ minimal over $(0 : a)$, then $\mathfrak{q}$ is minimal over $S^{-1}(0 : a) = (0 : a/1)$; for "$\Rightarrow$": every element of $S^{-1}A$ has the form a/s and its annihilator is $S^{-1}(0 : a)$). So $D(S^{-1}A) = D(A) \cap \mathrm{Spec}(S^{-1}A)$.

Decomposable zero ideal. Let $0 = \bigcap \mathfrak{q}_i$ be minimal, $\mathfrak{p}_i = r(\mathfrak{q}_i)$. ($\supseteq$) By 4.5 and its remark, $\mathfrak{p}_i = r(0 : x_i)$ with $(0 : x_i)$ $\mathfrak{p}_i$-primary; so $\mathfrak{p}_i$ is the unique minimal prime over $(0 : x_i)$, hence $\mathfrak{p}_i \in D(A)$. ($\subseteq$) Let $\mathfrak{p} \in D(A)$ be minimal over $(0 : a)$, $a \neq 0$. As in the proof of 4.5, $r(0 : a) = \bigcap_{a \notin \mathfrak{q}_i} \mathfrak{p}_i$ (since $(0 : a) = \bigcap (\mathfrak{q}_i : a)$, which is (1) if $a \in \mathfrak{q}_i$ and has radical $\mathfrak{p}_i$ otherwise). The prime $\mathfrak{p} \supseteq (0 : a)$ contains this intersection, hence some $\mathfrak{p}_i$ with $a \notin \mathfrak{q}_i$, and $\mathfrak{p}_i \supseteq r(0 : a) \supseteq (0 : a)$. By minimality of $\mathfrak{p}$, $\mathfrak{p} = \mathfrak{p}_i$. □

Exercise 10

Problem. For any prime ideal $\mathfrak{p}$ in a ring A, let $S_\mathfrak{p}(0)$ denote the kernel of the homomorphism $A \to A_\mathfrak{p}$. Prove that

(i) $S_\mathfrak{p}(0) \subseteq \mathfrak{p}$.

(ii) $r(S_\mathfrak{p}(0)) = \mathfrak{p} \iff \mathfrak{p}$ is a minimal prime ideal of A.

(iii) If $\mathfrak{p} \supseteq \mathfrak{p}'$, then $S_\mathfrak{p}(0) \subseteq S_{\mathfrak{p}'}(0)$.

(iv) $\bigcap_{\mathfrak{p} \in D(A)} S_\mathfrak{p}(0) = 0$, where $D(A)$ is defined in Exercise 9.

Solution. (i) If $x/1 = 0$ in $A_\mathfrak{p}$, then $sx = 0 \in \mathfrak{p}$ for some $s \notin \mathfrak{p}$, so $x \in \mathfrak{p}$.

(ii) $S_\mathfrak{p}(0)$ is the contraction of $0 \subseteq A_\mathfrak{p}$, and $r(0) = \mathfrak{N}(A_\mathfrak{p}) = \bigcap_{\mathfrak{p}' \subseteq \mathfrak{p}} \mathfrak{p}' A_\mathfrak{p}$ (3.13, 3.12); contracting (1.18: $r(\mathfrak{b})^c = r(\mathfrak{b}^c)$): $r(S_\mathfrak{p}(0)) = \bigcap_{\mathfrak{p}' \subseteq \mathfrak{p}} \mathfrak{p}'$. This equals $\mathfrak{p}$ iff $\mathfrak{p} \subseteq \mathfrak{p}'$ for every prime $\mathfrak{p}' \subseteq \mathfrak{p}$, i.e. iff $\mathfrak{p}$ is minimal.

(iii) If $sx = 0$ with $s \notin \mathfrak{p}$, then $s \notin \mathfrak{p}'$ (as $\mathfrak{p}' \subseteq \mathfrak{p}$), so $x/1 = 0$ in $A_{\mathfrak{p}'}$.

(iv) Let $x \neq 0$, so $(0 : x) \neq (1)$, and let $\mathfrak{p}$ be a prime minimal over $(0 : x)$; then $\mathfrak{p} \in D(A)$. If $x \in S_\mathfrak{p}(0)$ there is $s \notin \mathfrak{p}$ with $sx = 0$, i.e. $s \in (0 : x) \subseteq \mathfrak{p}$, a contradiction. So $x \notin \bigcap_{D(A)} S_\mathfrak{p}(0)$. □

Exercise 11

Problem. If $\mathfrak{p}$ is a minimal prime ideal of a ring A, show that $S_\mathfrak{p}(0)$ (Exercise 10) is the smallest $\mathfrak{p}$-primary ideal.

Let $\mathfrak{a}$ be the intersection of the ideals $S_\mathfrak{p}(0)$ as $\mathfrak{p}$ runs through the minimal prime ideals of A. Show that $\mathfrak{a}$ is contained in the nilradical of A.

Suppose that the zero ideal is decomposable. Prove that $\mathfrak{a} = 0$ if and only if every prime ideal of 0 is isolated.

Solution. *$S_\mathfrak{p}(0)$ is $\mathfrak{p}$-primary.* $A_\mathfrak{p}$ has a unique prime ideal $\mathfrak{p}A_\mathfrak{p}$ (3.13), which is therefore its nilradical. In $A_\mathfrak{p}$ every non-unit is nilpotent, so every zero-divisor is nilpotent and 0 is a primary ideal of $A_\mathfrak{p}$, with radical $\mathfrak{p}A_\mathfrak{p}$. Its contraction $S_\mathfrak{p}(0)$ is primary with radical $(\mathfrak{p}A_\mathfrak{p})^c = \mathfrak{p}$.

Smallest. If $\mathfrak{q}$ is $\mathfrak{p}$-primary, then $S_\mathfrak{p}(\mathfrak{q}) = \mathfrak{q}$ by 4.8(ii) (as $(A - \mathfrak{p}) \cap \mathfrak{p} = \emptyset$), and $S_\mathfrak{p}(0) \subseteq S_\mathfrak{p}(\mathfrak{q}) = \mathfrak{q}$.

$\mathfrak{a} \subseteq \mathfrak{N}$. By Ex. 10(i) $\mathfrak{a} \subseteq \bigcap_{\mathfrak{p}\ \min} \mathfrak{p} = \mathfrak{N}$ (every prime contains a minimal prime).

Decomposable zero ideal. Let $0 = \bigcap \mathfrak{q}_i$ be minimal, $\mathfrak{p}_i = r(\mathfrak{q}_i)$. Minimal primes of A are the minimal $\mathfrak{p}_i$ (4.6). For $\mathfrak{p}_i$ minimal, $S = A - \mathfrak{p}_i$ meets $\mathfrak{p}_j$ iff $\mathfrak{p}_j \not\subseteq \mathfrak{p}_i$ iff $j \neq i$ (by minimality, $\mathfrak{p}_j \subseteq \mathfrak{p}_i \Rightarrow \mathfrak{p}_j = \mathfrak{p}_i$), so by 4.9 $S_{\mathfrak{p}_i}(0) = \mathfrak{q}_i$. Hence $\mathfrak{a} = \bigcap_{\mathfrak{p}_i\ \min} \mathfrak{q}_i$. If every $\mathfrak{p}_i$ is isolated, then $\mathfrak{a} = \bigcap_{\text{all } i} \mathfrak{q}_i = 0$. Conversely if $\mathfrak{a} = 0$, then $0 = \bigcap_{\mathfrak{p}_i\ \min} \mathfrak{q}_i$ is a primary decomposition which (after removing redundant terms) is minimal and has only minimal primes as associated primes; by 4.5 the associated primes of 0 do not depend on the decomposition, so there are no embedded primes. □

Exercise 12

Problem. Let A be a ring, S a multiplicatively closed subset of A. For any ideal $\mathfrak{a}$, let $S(\mathfrak{a})$ denote the contraction in A of $S^{-1}\mathfrak{a}$. The ideal $S(\mathfrak{a})$ is called the *saturation* of $\mathfrak{a}$ with respect to S. Prove that

(i) $S(\mathfrak{a}) \cap S(\mathfrak{b}) = S(\mathfrak{a} \cap \mathfrak{b})$;

(ii) $S(r(\mathfrak{a})) = r(S(\mathfrak{a}))$;

(iii) $S(\mathfrak{a}) = (1) \iff \mathfrak{a}$ meets S;

(iv) $S_1(S_2(\mathfrak{a})) = (S_1 S_2)(\mathfrak{a})$.

If $\mathfrak{a}$ has a primary decomposition, prove that the set of ideals $S(\mathfrak{a})$ (where S runs through all multiplicatively closed subsets of A) is finite.

Solution. Recall $S(\mathfrak{a}) = \{x : sx \in \mathfrak{a} \text{ for some } s \in S\}$ (3.11(ii)).

(i) $S^{-1}(\mathfrak{a} \cap \mathfrak{b}) = S^{-1}\mathfrak{a} \cap S^{-1}\mathfrak{b}$ (3.4) and contraction commutes with intersection.

(ii) $S^{-1}r(\mathfrak{a}) = r(S^{-1}\mathfrak{a})$ (3.11(v)), and $r(\mathfrak{c})^c = r(\mathfrak{c}^c)$ (1.18); apply with $\mathfrak{c} = S^{-1}\mathfrak{a}$.

(iii) $S(\mathfrak{a}) = (1) \iff 1 \in S(\mathfrak{a}) \iff 1/1 \in S^{-1}\mathfrak{a} \iff S^{-1}\mathfrak{a} = (1) \iff \mathfrak{a} \cap S \neq \emptyset$ (3.11(ii)).

(iv) $x \in S_1(S_2(\mathfrak{a})) \iff s_1 x \in S_2(\mathfrak{a})$ for some $s_1 \in S_1 \iff s_2 s_1 x \in \mathfrak{a}$ for some $s_1 \in S_1$, $s_2 \in S_2 \iff x \in (S_1 S_2)(\mathfrak{a})$.

Finiteness. Let $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$ be a minimal decomposition, $\mathfrak{p}_i = r(\mathfrak{q}_i)$. By 4.9, $S(\mathfrak{a}) = \bigcap_{i:\mathfrak{p}_i \cap S = \emptyset} \mathfrak{q}_i$ (the empty intersection being (1)). So $S(\mathfrak{a})$ depends only on the subset $\{i : \mathfrak{p}_i \cap S = \emptyset\} \subseteq \{1, \ldots, n\}$; there are at most 2^n such ideals. □

Exercise 13

Problem. Let A be a ring and $\mathfrak{p}$ a prime ideal of A. The nth *symbolic power* of $\mathfrak{p}$ is defined to be the ideal (in the notation of Exercise 12) $\mathfrak{p}^{(n)} = S_{\mathfrak{p}}(\mathfrak{p}^n)$, where $S_{\mathfrak{p}} = A - \mathfrak{p}$. Show that

(i) $\mathfrak{p}^{(n)}$ is a $\mathfrak{p}$-primary ideal;

(ii) if $\mathfrak{p}^n$ has a primary decomposition, then $\mathfrak{p}^{(n)}$ is its $\mathfrak{p}$-primary component;

(iii) if $\mathfrak{p}^{(m)}\mathfrak{p}^{(n)}$ has a primary decomposition, then $\mathfrak{p}^{(m+n)}$ is its $\mathfrak{p}$-primary component;

(iv) $\mathfrak{p}^{(n)} = \mathfrak{p}^n \iff \mathfrak{p}^n$ is $\mathfrak{p}$-primary.

Solution. (i) In $A_{\mathfrak{p}}$, $r(\mathfrak{p}^n A_{\mathfrak{p}}) = \mathfrak{p}A_{\mathfrak{p}}$ is maximal, so $\mathfrak{p}^n A_{\mathfrak{p}}$ is primary (4.2); its contraction $\mathfrak{p}^{(n)}$ is primary with radical $(\mathfrak{p}A_{\mathfrak{p}})^c = \mathfrak{p}$.

(ii) $r(\mathfrak{p}^n) = \mathfrak{p}$ (1.13(vi)); every prime containing $\mathfrak{p}^n$ contains $\mathfrak{p}$, so $\mathfrak{p}$ is the unique minimal prime of $\mathfrak{p}^n$, and the $\mathfrak{p}$-component of any minimal decomposition is isolated. By 4.9/4.11 the isolated component is $S(\mathfrak{p}^n)$ for $S = A - \mathfrak{p}$, i.e. $\mathfrak{p}^{(n)}$.

(iii) Put $\mathfrak{b} = \mathfrak{p}^{(m)}\mathfrak{p}^{(n)}$. $r(\mathfrak{b}) = r(\mathfrak{p}^{(m)}) \cap r(\mathfrak{p}^{(n)}) = \mathfrak{p}$ (1.13), so as in (ii), $\mathfrak{p}$ is the unique minimal prime of $\mathfrak{b}$ and its component is $S_{\mathfrak{p}}(\mathfrak{b})$. Now $S_{\mathfrak{p}}^{-1}\mathfrak{p}^{(m)} = \mathfrak{p}^m A_{\mathfrak{p}}$ (since $\mathfrak{p}^m A_{\mathfrak{p}}$ is an extended ideal, $(\mathfrak{p}^m A_{\mathfrak{p}})^{ce} = \mathfrak{p}^m A_{\mathfrak{p}}$), so $S_{\mathfrak{p}}^{-1}\mathfrak{b} = \mathfrak{p}^m A_{\mathfrak{p}} \cdot \mathfrak{p}^n A_{\mathfrak{p}} = \mathfrak{p}^{m+n} A_{\mathfrak{p}}$ and $S_{\mathfrak{p}}(\mathfrak{b}) = \mathfrak{p}^{(m+n)}$.

(iv) ($\Leftarrow$) If $\mathfrak{p}^n$ is $\mathfrak{p}$-primary, then $S_{\mathfrak{p}}(\mathfrak{p}^n) = \mathfrak{p}^n$ by 4.8(ii). ($\Rightarrow$) $\mathfrak{p}^n = \mathfrak{p}^{(n)}$ is $\mathfrak{p}$-primary by (i). □

Exercise 14

Problem. Let $\mathfrak{a}$ be a decomposable ideal in a ring A and let $\mathfrak{p}$ be a maximal element of the set of ideals $(\mathfrak{a} : x)$, where $x \in A$ and $x \notin \mathfrak{a}$. Show that $\mathfrak{p}$ is a prime ideal belonging to $\mathfrak{a}$.

Solution. Let $\mathfrak{p} = (\mathfrak{a} : x)$ with $x \notin \mathfrak{a}$ (so $\mathfrak{p} \neq (1)$) maximal in the set. Suppose $yz \in \mathfrak{p}$ with $z \notin \mathfrak{p}$. Then $yzx \in \mathfrak{a}$, so $y \in (\mathfrak{a} : zx)$. Also $zx \notin \mathfrak{a}$ (else $z \in (\mathfrak{a} : x) = \mathfrak{p}$) and $(\mathfrak{a} : zx) \supseteq (\mathfrak{a} : x) = \mathfrak{p}$. By maximality $(\mathfrak{a} : zx) = \mathfrak{p}$, so $y \in \mathfrak{p}$. Hence $\mathfrak{p}$ is prime. Then $r(\mathfrak{a} : x) = \mathfrak{p}$ is a prime of the form $r(\mathfrak{a} : x)$, so by the first uniqueness theorem (4.5) $\mathfrak{p}$ is one of the primes belonging to $\mathfrak{a}$. □

Exercise 15

Problem. Let $\mathfrak{a}$ be a decomposable ideal in a ring A and let Σ be an isolated set of prime ideals belonging to $\mathfrak{a}$, and let $\mathfrak{q}_\Sigma$ be the intersection of the corresponding primary components. Let f be an element of A such that, for each prime ideal $\mathfrak{p}$ belonging to $\mathfrak{a}$, we have $f \in \mathfrak{p} \iff \mathfrak{p} \notin \Sigma$, and let S_f be the set of all powers of f. Show that $\mathfrak{q}_\Sigma = S_f(\mathfrak{a}) = (\mathfrak{a} : f^n)$ for all large n.

Solution. Let $\mathfrak{a} = \bigcap \mathfrak{q}_i$ be minimal, $\mathfrak{p}_i = r(\mathfrak{q}_i)$. For $S = S_f$: $S \cap \mathfrak{p}_i \neq \emptyset \iff f \in \mathfrak{p}_i \iff \mathfrak{p}_i \notin \Sigma$. By 4.9, $S_f(\mathfrak{a}) = \bigcap_{\mathfrak{p}_i \in \Sigma} \mathfrak{q}_i = \mathfrak{q}_\Sigma$.

Moreover $(\mathfrak{a} : f^n) = \bigcap_i (\mathfrak{q}_i : f^n)$. If $\mathfrak{p}_i \notin \Sigma$, then $f \in \mathfrak{p}_i = r(\mathfrak{q}_i)$, so $f^n \in \mathfrak{q}_i$ for n large and $(\mathfrak{q}_i : f^n) = (1)$. If $\mathfrak{p}_i \in \Sigma$, then $f^n \notin \mathfrak{p}_i$ and $(\mathfrak{q}_i : f^n) = \mathfrak{q}_i$ by 4.4(iii). Taking n large for all of the finitely many i, $(\mathfrak{a} : f^n) = \bigcap_{\mathfrak{p}_i \in \Sigma} \mathfrak{q}_i = \mathfrak{q}_\Sigma$. □

Exercise 16

Problem. If A is a ring in which every ideal has a primary decomposition, show that every ring of fractions $S^{-1}A$ has the same property.

Solution. Let $\mathfrak{b}$ be an ideal of $S^{-1}A$. By 3.11(i), $\mathfrak{b} = \mathfrak{a}^e = S^{-1}\mathfrak{a}$ with $\mathfrak{a} = \mathfrak{b}^c$. Write $\mathfrak{a} = \bigcap_i \mathfrak{q}_i$ (primary). By 3.4, $S^{-1}\mathfrak{a} = \bigcap_i S^{-1}\mathfrak{q}_i$. By 4.8, each $S^{-1}\mathfrak{q}_i$ is either $S^{-1}A$ (if $S \cap r(\mathfrak{q}_i) \neq \emptyset$) or primary in $S^{-1}A$. Discarding the components equal to the unit ideal, $\mathfrak{b}$ is a finite intersection of primary ideals (if all components are the unit ideal, $\mathfrak{b} = S^{-1}A$, the empty intersection). □

Exercise 17

Problem. Let A be a ring with the following property.

(L1) For every ideal $\mathfrak{a} \neq (1)$ in A and every prime ideal $\mathfrak{p}$, there exists $x \notin \mathfrak{p}$ such that $S_\mathfrak{p}(\mathfrak{a}) = (\mathfrak{a} : x)$, where $S_\mathfrak{p} = A - \mathfrak{p}$.

Then every ideal in A is an intersection of (possibly infinitely many) primary ideals. [Let $\mathfrak{a}$ be an ideal $\neq (1)$ in A, and let $\mathfrak{p}_1$ be a minimal element of the set of prime ideals containing $\mathfrak{a}$. Then $\mathfrak{q}_1 = S_{\mathfrak{p}_1}(\mathfrak{a})$ is $\mathfrak{p}_1$-primary (by Exercise 11), and $\mathfrak{q}_1 = (\mathfrak{a} : x)$ for some $x \notin \mathfrak{p}_1$. Show that $\mathfrak{a} = \mathfrak{q}_1 \cap (\mathfrak{a} + (x))$. Now let $\mathfrak{a}_1$ be a maximal element of the set of ideals $\mathfrak{b} \supseteq \mathfrak{a}$ such that $\mathfrak{q}_1 \cap \mathfrak{b} = \mathfrak{a}$, and choose $\mathfrak{a}_1$ so that $x \in \mathfrak{a}_1$, and therefore $\mathfrak{a}_1 \not\subseteq \mathfrak{p}_1$. Repeat the construction starting with $\mathfrak{a}_1$, and so on. At the nth stage we have $\mathfrak{a} = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_n \cap \mathfrak{a}_n$ where the $\mathfrak{q}_i$ are primary ideals, $\mathfrak{a}_n$ is maximal among the ideals $\mathfrak{b} \supseteq \mathfrak{a}_{n-1} = \mathfrak{a}_n \cap \mathfrak{q}_n$ such that $\mathfrak{a} = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_n \cap \mathfrak{b}$, and $\mathfrak{a}_n \not\subseteq \mathfrak{p}_n$. If at any stage we have $\mathfrak{a}_n = (1)$, the process stops, and $\mathfrak{a}$ is a finite intersection of primary ideals. If not, continue by transfinite induction, observing that each $\mathfrak{a}_n$ strictly contains $\mathfrak{a}_{n-1}$.]

Solution. Let $\mathfrak{a} \neq (1)$. We construct, by transfinite recursion, ideals $\mathfrak{a}_\alpha$ (with $\mathfrak{a}_0 = \mathfrak{a}$) increasing strictly with α, and primary ideals $\mathfrak{q}_\alpha$ ($\alpha \geq 1$), such that

$$(*_\alpha) \qquad \mathfrak{a} = \bigcap_{1 \leq \beta \leq \alpha} \mathfrak{q}_\beta \ \cap \ \mathfrak{a}_\alpha.$$

Key step. Suppose $(*_\alpha)$ holds and $\mathfrak{a}_\alpha \neq (1)$. Let $\mathfrak{p}$ be a prime minimal over $\mathfrak{a}_\alpha$. Passing to $A/\mathfrak{a}_\alpha$, Ex. 11 shows $S_\mathfrak{p}(\mathfrak{a}_\alpha)$ (the preimage of $S_{\bar{\mathfrak{p}}}(0)$) is a $\mathfrak{p}$-primary ideal $\mathfrak{q}$. By (L1), $\mathfrak{q} = (\mathfrak{a}_\alpha : x)$ for some $x \notin \mathfrak{p}$. We claim $\mathfrak{a}_\alpha = \mathfrak{q} \cap (\mathfrak{a}_\alpha + (x))$. The inclusion $\subseteq$ is clear. Let $y = a' + cx \in \mathfrak{q}$ with $a' \in \mathfrak{a}_\alpha$. Then $yx \in \mathfrak{a}_\alpha$, so $cx^2 \in \mathfrak{a}_\alpha$, i.e. $c \in (\mathfrak{a}_\alpha : x^2) \subseteq S_\mathfrak{p}(\mathfrak{a}_\alpha) = (\mathfrak{a}_\alpha : x)$ (as $x^2 \notin \mathfrak{p}$). Hence $cx \in \mathfrak{a}_\alpha$ and $y \in \mathfrak{a}_\alpha$.

Successor stage. Put $\mathfrak{q}_{\alpha+1} = \mathfrak{q}$ and $x_{\alpha+1} = x$. Let M be the set of ideals $\mathfrak{b} \supseteq \mathfrak{a}_\alpha + (x)$ with $\mathfrak{q} \cap \mathfrak{b} = \mathfrak{a}_\alpha$; it contains $\mathfrak{a}_\alpha + (x)$, and the union of a chain in M lies in M, so by Zorn M has a maximal element $\mathfrak{a}_{\alpha+1}$. Then $x \in \mathfrak{a}_{\alpha+1}$, $\mathfrak{a}_\alpha \subseteq \mathfrak{p}$ and $x \notin \mathfrak{p}$ give $\mathfrak{a}_{\alpha+1} \supsetneq \mathfrak{a}_\alpha$, and $(*_{\alpha+1})$ holds since $\mathfrak{a} = \bigcap_{\beta \leq \alpha} \mathfrak{q}_\beta \cap (\mathfrak{q} \cap \mathfrak{a}_{\alpha+1})$. (Note $\mathfrak{a}_{\alpha+1} \not\subseteq \mathfrak{p}$.)

Limit stage λ. Put $\mathfrak{a}_\lambda = \bigcup_{\beta<\lambda} \mathfrak{a}_\beta$. If $y \in \bigcap_{\beta<\lambda} \mathfrak{q}_\beta \cap \mathfrak{a}_\lambda$ then $y \in \mathfrak{a}_\beta$ for some $\beta < \lambda$, so $y \in \bigcap_{\gamma \leq \beta} \mathfrak{q}_\gamma \cap \mathfrak{a}_\beta = \mathfrak{a}$; hence $\mathfrak{a} = \bigcap_{\beta<\lambda} \mathfrak{q}_\beta \cap \mathfrak{a}_\lambda$.

Termination. The $\mathfrak{a}_\alpha$ are strictly increasing, and there is only a set of ideals, so at some stage $\mathfrak{a}_\alpha = (1)$ (a limit stage has $\mathfrak{a}_\lambda \neq (1)$ only if the process continues). Then $(*_\alpha)$ reads $\mathfrak{a} = \bigcap_{\beta \leq \alpha} \mathfrak{q}_\beta$, an intersection of primary ideals. □

Exercise 18

Problem. Consider the following condition on a ring A:

(L2) Given an ideal $\mathfrak{a}$ and a descending chain $S_1 \supseteq S_2 \supseteq \cdots \supseteq S_n \supseteq \cdots$ of multiplicatively closed subsets of A, there exists an integer n such that $S_n(\mathfrak{a}) = S_{n+1}(\mathfrak{a}) = \cdots$.

Prove that the following are equivalent:

(i) Every ideal in A has a primary decomposition.

(ii) A satisfies (L1) and (L2).

[For (i)⇒(ii), use Exercises 12 and 15. For (ii)⇒(i) show, with the notation of the proof of Exercise 17, that if $S_n = S_{\mathfrak{p}_1} \cap \cdots \cap S_{\mathfrak{p}_n}$ then S_n meets $\mathfrak{a}_n$, hence $S_n(\mathfrak{a}_n) = (1)$, and therefore $S_n(\mathfrak{a}) = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_n$. Now use (L2) to show that the construction must terminate after a finite number of steps.]

Solution. (i)⇒(ii). *(L2).* If $S_1 \supseteq S_2 \supseteq \cdots$ then $S_1(\mathfrak{a}) \supseteq S_2(\mathfrak{a}) \supseteq \cdots$ (since $S(\mathfrak{a}) = \bigcup_{s \in S}(\mathfrak{a} : s)$), and by Ex. 12 the set of all ideals $S(\mathfrak{a})$ is finite, so the chain stabilizes.

(L1). Let $\mathfrak{a} \neq (1)$ and $\mathfrak{p}$ prime. Let $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$ be a minimal decomposition, $\mathfrak{p}_i = r(\mathfrak{q}_i)$, and $\Sigma = \{\mathfrak{p}_i : \mathfrak{p}_i \subseteq \mathfrak{p}\}$. Σ is isolated (if $\mathfrak{p}_k \subseteq \mathfrak{p}_j \in \Sigma$ then $\mathfrak{p}_k \subseteq \mathfrak{p}$). Let $\mathfrak{b} = \bigcap_{\mathfrak{p}_i \notin \Sigma} \mathfrak{p}_i$ (= A if Σ is everything). We claim $\mathfrak{b} \not\subseteq \mathfrak{p}$ and $\mathfrak{b} \not\subseteq \mathfrak{p}_j$ for $\mathfrak{p}_j \in \Sigma$: if $\mathfrak{b} \subseteq \mathfrak{p}$ (resp. $\mathfrak{p}_j$), then since $\prod_{\mathfrak{p}_i \notin \Sigma} \mathfrak{p}_i \subseteq \mathfrak{b}$ and $\mathfrak{p}$ is prime, some $\mathfrak{p}_i \notin \Sigma$ satisfies $\mathfrak{p}_i \subseteq \mathfrak{p}$ (resp. $\mathfrak{p}_i \subseteq \mathfrak{p}_j \subseteq \mathfrak{p}$), contradicting $\mathfrak{p}_i \notin \Sigma$. By prime avoidance (1.11) there is $f \in \mathfrak{b}$ with $f \notin \mathfrak{p}$ and $f \notin \mathfrak{p}_j$ for all $\mathfrak{p}_j \in \Sigma$. Then for each $\mathfrak{p}_i$: $f \in \mathfrak{p}_i \iff \mathfrak{p}_i \notin \Sigma$. By Ex. 15, $(\mathfrak{a} : f^n) = \mathfrak{q}_\Sigma$ for n large. By 4.9 applied to $S = A - \mathfrak{p}$ (which meets $\mathfrak{p}_i$ iff $\mathfrak{p}_i \not\subseteq \mathfrak{p}$), $S_\mathfrak{p}(\mathfrak{a}) = \bigcap_{\mathfrak{p}_i \subseteq \mathfrak{p}} \mathfrak{q}_i = \mathfrak{q}_\Sigma$. So $x = f^n \notin \mathfrak{p}$ satisfies $S_\mathfrak{p}(\mathfrak{a}) = (\mathfrak{a} : x)$.

(ii)⇒(i). Let $\mathfrak{a} \neq (1)$ and run the construction of Ex. 17 (with the following precise choice in the successor step: $\mathfrak{a}_n$ is maximal in $M_n = \{\mathfrak{b} \supseteq \mathfrak{a}_{n-1} + (x_n) : \ \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_n \cap \mathfrak{b} = \mathfrak{a}\}$ — this set contains $\mathfrak{a}_{n-1}+(x_n)$ because $\mathfrak{q}_n \cap (\mathfrak{a}_{n-1}+(x_n)) = \mathfrak{a}_{n-1}$). Write $Q_n = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_n$, so $\mathfrak{a} = Q_n \cap \mathfrak{a}_n$, with $\mathfrak{q}_i$ being $\mathfrak{p}_i$-primary and $x_i \in \mathfrak{a}_i$, $x_i \notin \mathfrak{p}_i$, $\mathfrak{a}_{i-1} \subseteq \mathfrak{p}_i$.

Let $S_n = S_{\mathfrak{p}_1} \cap \cdots \cap S_{\mathfrak{p}_n} = A - (\mathfrak{p}_1 \cup \cdots \cup \mathfrak{p}_n)$; these form a descending chain. $\mathfrak{a}_n \not\subseteq \mathfrak{p}_i$ for $i \leq n$ (since $x_i \in \mathfrak{a}_i \subseteq \mathfrak{a}_n$, $x_i \notin \mathfrak{p}_i$), so by prime avoidance $\mathfrak{a}_n \not\subseteq \bigcup_{i \leq n} \mathfrak{p}_i$, i.e. S_n meets $\mathfrak{a}_n$ and $S_n(\mathfrak{a}_n) = (1)$ (Ex. 12(iii)). Also $S_n \cap \mathfrak{p}_i = \emptyset$ for $i \leq n$, so $S_n(\mathfrak{q}_i) = \mathfrak{q}_i$ (4.8). Hence by Ex. 12(i), $S_n(\mathfrak{a}) = S_n(Q_n) \cap S_n(\mathfrak{a}_n) = Q_n$.

Suppose the process never stops at a finite stage, i.e. $\mathfrak{a}_n \neq (1)$ for all n. By (L2), $S_N(\mathfrak{a}) = S_{N+1}(\mathfrak{a})$ for some N, so $Q_N = Q_{N+1} = Q_N \cap \mathfrak{q}_{N+1}$, whence $Q_N \subseteq \mathfrak{q}_{N+1}$. Then $\mathfrak{a} = Q_{N+1} \cap \mathfrak{a}_{N+1} = Q_N \cap \mathfrak{a}_{N+1}$, and $\mathfrak{a}_{N+1} \supseteq \mathfrak{a}_N \supseteq \mathfrak{a}_{N-1} + (x_N)$. So $\mathfrak{a}_{N+1} \in M_N$, and maximality of $\mathfrak{a}_N$ in M_N forces $\mathfrak{a}_{N+1} = \mathfrak{a}_N$. But $x_{N+1} \in \mathfrak{a}_{N+1}$ while $x_{N+1} \notin \mathfrak{p}_{N+1} \supseteq \mathfrak{a}_N$, so $\mathfrak{a}_{N+1} \neq \mathfrak{a}_N$: contradiction. Hence $\mathfrak{a}_n = (1)$ for some finite n, and then $\mathfrak{a} = Q_n = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_n$ is a primary decomposition. □

Exercise 19

> **Problem.** Let A be a ring and let $\mathfrak{p}$ be a prime ideal of A. Show that every $\mathfrak{p}$-primary ideal contains $S_{\mathfrak{p}}(0)$, the kernel of the canonical homomorphism $A \to A_{\mathfrak{p}}$.
>
> Suppose that A satisfies the following condition: for every prime ideal $\mathfrak{p}$, the intersection of all $\mathfrak{p}$-primary ideals of A is equal to $S_{\mathfrak{p}}(0)$. (Noetherian rings satisfy this condition: see Chapter 10.) Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_n$ be distinct prime ideals, none of which is a minimal prime ideal of A. Then there exists an ideal $\mathfrak{a}$ in A whose associated prime ideals are $\mathfrak{p}_1, \ldots, \mathfrak{p}_n$.
>
> [Proof by induction on n. The case $n = 1$ is trivial (take $\mathfrak{a} = \mathfrak{p}_1$). Suppose $n > 1$ and let $\mathfrak{p}_n$ be maximal in the set $\{\mathfrak{p}_1, \ldots, \mathfrak{p}_n\}$. By the inductive hypothesis there exists an ideal $\mathfrak{b}$ and a minimal primary decomposition $\mathfrak{b} = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_{n-1}$, where each $\mathfrak{q}_i$ is $\mathfrak{p}_i$-primary. If $\mathfrak{b} \subseteq S_{\mathfrak{p}_n}(0)$, let $\mathfrak{p}$ be a minimal prime ideal of A contained in $\mathfrak{p}_n$. Then $S_{\mathfrak{p}_n}(0) \subseteq S_{\mathfrak{p}}(0)$, hence $\mathfrak{b} \subseteq S_{\mathfrak{p}}(0)$. Taking radicals and using Exercise 10, we have $\mathfrak{p}_1 \cap \cdots \cap \mathfrak{p}_{n-1} \subseteq \mathfrak{p}$, hence some $\mathfrak{p}_i \subseteq \mathfrak{p}$, hence $\mathfrak{p}_i = \mathfrak{p}$ since $\mathfrak{p}$ is minimal. This is a contradiction since no $\mathfrak{p}_i$ is minimal. Hence $\mathfrak{b} \not\subseteq S_{\mathfrak{p}_n}(0)$ and therefore there exists a $\mathfrak{p}_n$-primary ideal $\mathfrak{q}_n$ such that $\mathfrak{b} \not\subseteq \mathfrak{q}_n$. Show that $\mathfrak{a} = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_n$ has the required properties.]

Solution. *Every $\mathfrak{p}$-primary $\mathfrak{q}$ contains $S_{\mathfrak{p}}(0)$:* $S_{\mathfrak{p}}(\mathfrak{q}) = \mathfrak{q}$ by 4.8(ii), and $S_{\mathfrak{p}}(0) \subseteq S_{\mathfrak{p}}(\mathfrak{q})$.

The construction. Induct on n as in the hint; the case $n = 1$ is $\mathfrak{a} = \mathfrak{p}_1$ ($\mathfrak{p}_1$ is primary). Let $n > 1$ and $\mathfrak{p}_n$ maximal in $\{\mathfrak{p}_1, \ldots, \mathfrak{p}_n\}$. By induction there is $\mathfrak{b} = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_{n-1}$, minimal, with $\mathfrak{q}_i$ $\mathfrak{p}_i$-primary. The hint shows $\mathfrak{b} \not\subseteq S_{\mathfrak{p}_n}(0)$: if $\mathfrak{b} \subseteq S_{\mathfrak{p}_n}(0)$, take a minimal prime $\mathfrak{p} \subseteq \mathfrak{p}_n$ of A; by Ex. 10(iii) $S_{\mathfrak{p}_n}(0) \subseteq S_{\mathfrak{p}}(0)$, so $\mathfrak{b} \subseteq S_{\mathfrak{p}}(0)$; taking radicals, $\mathfrak{p}_1 \cap \cdots \cap \mathfrak{p}_{n-1} = r(\mathfrak{b}) \subseteq r(S_{\mathfrak{p}}(0)) = \mathfrak{p}$ (Ex. 10(ii)), so some $\mathfrak{p}_i \subseteq \mathfrak{p}$ by 1.11, whence $\mathfrak{p}_i = \mathfrak{p}$ is minimal, contradicting the hypothesis. By the assumed condition, $S_{\mathfrak{p}_n}(0) = \bigcap\{\mathfrak{q} : \ \mathfrak{q}\ \mathfrak{p}_n\text{-primary}\}$, so there is a $\mathfrak{p}_n$-primary $\mathfrak{q}_n$ with $\mathfrak{b} \not\subseteq \mathfrak{q}_n$.

Put $\mathfrak{a} = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_n = \mathfrak{b} \cap \mathfrak{q}_n$. The radicals $\mathfrak{p}_1, \ldots, \mathfrak{p}_n$ are distinct. *Irredundance.* For $i = n$: $\bigcap_{j<n} \mathfrak{q}_j = \mathfrak{b} \not\subseteq \mathfrak{q}_n$. For $i < n$: by minimality of $\mathfrak{b}$ there is $y \in \bigcap_{j<n, j\neq i} \mathfrak{q}_j$ with $y \notin \mathfrak{q}_i$. Since $\mathfrak{p}_n$ is maximal in the set and $\mathfrak{p}_i \neq \mathfrak{p}_n$, $\mathfrak{p}_n \not\subseteq \mathfrak{p}_i$; pick $s \in \mathfrak{p}_n - \mathfrak{p}_i$ and k with $s^k \in \mathfrak{q}_n$. Then $ys^k \in \bigcap_{j\neq i} \mathfrak{q}_j$, and $ys^k \notin \mathfrak{q}_i$ (otherwise, $\mathfrak{q}_i$ being primary and $y \notin \mathfrak{q}_i$, $s^{km} \in \mathfrak{q}_i$ for some m, so $s \in \mathfrak{p}_i$). So no component is redundant and the decomposition is minimal. By 4.5 the associated primes of $\mathfrak{a}$ are exactly $\mathfrak{p}_1, \ldots, \mathfrak{p}_n$. □

Exercise 20

Problem. (*Primary decomposition of modules.*) Let M be a fixed A-module, N a submodule of M. The *radical* of N in M is defined to be $r_M(N) = \{x \in A : x^q M \subseteq N \text{ for some } q > 0\}$. Show that $r_M(N) = r(N : M) = r(\mathrm{Ann}(M/N))$. In particular, $r_M(N)$ is an ideal. State and prove the formulas for r_M analogous to (1.13).

Solution. $(N : M) = \mathrm{Ann}(M/N)$ (Ex. 2.2(ii) with $P = M$), and $x \in r(N : M) \iff x^q \in (N : M) \iff x^q M \subseteq N$ for some q. So $r_M(N) = r(N : M) = r(\mathrm{Ann}(M/N))$, an ideal.

Analogues of 1.13. For submodules N, P of M:

(i) $r_M(N) \supseteq (N : M) \supseteq \mathrm{Ann}(M)$ (analogue of $r(\mathfrak{a}) \supseteq \mathfrak{a}$), and $r_M(N)$ is the set of elements of A nilpotent in M/N.

(ii) $r(r_M(N)) = r_M(N)$ (analogue of $r(r(\mathfrak{a})) = r(\mathfrak{a})$): it is the radical of the ideal $(N : M)$. Also $N \subseteq P \Rightarrow r_M(N) \subseteq r_M(P)$.

(iii) $r_M(N \cap P) = r_M(N) \cap r_M(P)$ (analogue of 1.13(iii)): $(N \cap P : M) = (N : M) \cap (P : M)$ and r commutes with finite intersections (1.13(iii)).

(iv) $r_M(N) = (1) \iff N = M$ (analogue of 1.13(iv)): $1 \in r_M(N) \iff M \subseteq N$.

(v) $r_M(N) = \bigcap\{\mathfrak{p} \text{ prime} : \mathfrak{p} \supseteq (N : M)\}$ (by 1.14); in particular $r_M(0) = r(\mathrm{Ann}\, M)$.

(vi) For an ideal $\mathfrak{a}$: $r_M(\mathfrak{a}M) \supseteq r(\mathfrak{a} + \mathrm{Ann}\, M)$ always, with equality if M is finitely generated (analogue of 1.13(vi)).

Proof of (vi). If $x^k \in \mathfrak{a} + \mathrm{Ann}\, M$ then $x^k M \subseteq \mathfrak{a}M$. Conversely let M be finitely generated and $x^q M \subseteq \mathfrak{a}M$. By Prop. 2.4 applied to $\phi = x^q$ and the ideal $\mathfrak{a}$, $x^{qn} + a_1 x^{q(n-1)} + \cdots + a_n$ annihilates M for some $n \geq 1$ and $a_i \in \mathfrak{a}$, so $x^{qn} \in \mathfrak{a} + \mathrm{Ann}\, M$. □

Exercise 21

Problem. An element $x \in A$ defines an endomorphism ϕ_x of M, namely $m \mapsto xm$. The element x is said to be a *zero-divisor* (resp. *nilpotent*) *in* M if ϕ_x is not injective (resp. is nilpotent). A submodule Q of M is *primary in* M if $Q \neq M$ and every zero-divisor in M/Q is nilpotent.

Show that if Q is primary in M, then $(Q : M)$ is a primary ideal and hence $r_M(Q)$ is a prime ideal $\mathfrak{p}$. We say that Q is $\mathfrak{p}$-primary (in M).

Prove the analogues of (4.3) and (4.4).

Solution. $(Q : M)$ *is primary.* $(Q : M) \neq (1)$ since $Q \neq M$. Suppose $xy \in (Q : M)$ and $x \notin (Q : M)$. Then there is $m \in M$ with $xm \notin Q$, while $y(xm) = (xy)m \in Q$. So y kills the non-zero class $xm + Q \in M/Q$, i.e. y is a zero-divisor in M/Q, hence nilpotent there: $y^q M \subseteq Q$, i.e. $y^q \in (Q : M)$. Thus $(Q : M)$ is primary, and $\mathfrak{p} = r_M(Q) = r(Q : M)$ is prime (4.1).

Analogue of 4.3. If $Q_1, \ldots, Q_n$ are $\mathfrak{p}$-primary in M, then $Q = \bigcap Q_i$ is $\mathfrak{p}$-primary in M. Proof: $Q \neq M$ as $Q \subseteq Q_1$; $r_M(Q) = \bigcap r_M(Q_i) = \mathfrak{p}$ by Ex. 20(iii). Let x be a zero-divisor in M/Q:

$xm \in Q$ for some $m \notin Q$. Pick i with $m \notin Q_i$; then $xm \in Q_i$, so x is a zero-divisor in M/Q_i, hence nilpotent there, so $x \in r_M(Q_i) = \mathfrak{p}$. Then $x^{q_j} M \subseteq Q_j$ for each j, and with $q = \max q_j$, $x^q M \subseteq Q$: x is nilpotent in M/Q.

Analogue of 4.4. Let Q be $\mathfrak{p}$-primary in M, $m \in M$, and $(Q : m) = \{a \in A : am \in Q\}$. Then:

(i) if $m \in Q$ then $(Q : m) = (1)$;

(ii) if $m \notin Q$ then $(Q : m)$ is $\mathfrak{p}$-primary (in A), hence $r(Q : m) = \mathfrak{p}$;

(iii) if $x \in A - \mathfrak{p}$ then $(Q :_M x) = \{m \in M : xm \in Q\} = Q$.

Proofs: (i) clear. (iii) $x \notin \mathfrak{p} = r_M(Q)$ means x is not nilpotent in M/Q, hence not a zero-divisor there (zero-divisors are nilpotent), so $xm \in Q \Rightarrow m \in Q$. (ii) $(Q : m) \neq (1)$. If $a \in (Q : m)$ then a kills the non-zero class $m + Q$, so a is a zero-divisor in M/Q, hence in $\mathfrak{p}$; so $(Q : M) \subseteq (Q : m) \subseteq \mathfrak{p}$, and taking radicals $r(Q : m) = \mathfrak{p}$. Primary: let $yz \in (Q : m)$ with $z \notin \mathfrak{p}$. By (iii)-reasoning z acts injectively on M/Q; since $z(ym) = (yz)m \in Q$, we get $ym \in Q$, i.e. $y \in (Q : m)$. If instead $z \in \mathfrak{p}$, then $z^q \in (Q : M) \subseteq (Q : m)$. So $(Q : m)$ is primary. □

Exercise 22

Problem. A *primary decomposition of N in M* is a representation of N as an intersection $N = Q_1 \cap \cdots \cap Q_n$ of primary submodules of M; it is a *minimal primary decomposition* if the ideals $\mathfrak{p}_i = r_M(Q_i)$ are all distinct and if none of the components Q_i can be omitted from the intersection, that is if $Q_i \not\supseteq \bigcap_{j \neq i} Q_j$ $(1 \leq i \leq n)$.

Prove the analogue of (4.5), that the prime ideals $\mathfrak{p}_i$ depend only on N (and M). They are called the *prime ideals belonging to N in M*. Show that they are also the prime ideals belonging to 0 in M/N.

Solution. Let $N = \bigcap Q_i$ be minimal, $\mathfrak{p}_i = r_M(Q_i)$. For $m \in M$, $(N : m) = \bigcap_i (Q_i : m)$ (Ex. 20), and by the analogue of 4.4, $(Q_i : m) = (1)$ if $m \in Q_i$ and $\mathfrak{p}_i$-primary (radical $\mathfrak{p}_i$) if not. Hence

$$r(N : m) = \bigcap_{i:\ m \notin Q_i} \mathfrak{p}_i.$$

Claim: $\{\mathfrak{p}_1, \ldots, \mathfrak{p}_n\}$ is the set of prime ideals of the form $r(N : m)$, $m \in M$. If $r(N : m)$ is prime, it equals $\mathfrak{p}_j$ for some j (a prime equal to a finite intersection of primes equals one of them: it contains the product, hence one $\mathfrak{p}_i$, and is contained in all of them). Conversely for each i, minimality gives $m_i \in \bigcap_{j \neq i} Q_j - Q_i$, and then $r(N : m_i) = \mathfrak{p}_i$. Thus the set $\{\mathfrak{p}_i\}$ is intrinsic to N and M.

Relation to M/N. Since $N \subseteq Q_i$, $0 = \bigcap (Q_i/N)$ in M/N. $(M/N)/(Q_i/N) \cong M/Q_i$, so Q_i/N is primary in M/N, and $r_{M/N}(Q_i/N) = r(\mathrm{Ann}(M/Q_i)) = r_M(Q_i) = \mathfrak{p}_i$. Irredundance of $\bigcap Q_i$ translates to that of $\bigcap Q_i/N$. So this is a minimal primary decomposition of 0 in M/N with the same radicals; the $\mathfrak{p}_i$ are the primes belonging to 0 in M/N. □

Exercise 23

Problem. State and prove the analogues of (4.6)–(4.11) inclusive. (There is no loss of generality in taking $N = 0$.)

Solution. Let M be an A-module in which 0 is decomposable: $0 = Q_1 \cap \cdots \cap Q_n$, a minimal primary decomposition, $\mathfrak{p}_i = r_M(Q_i)$. A prime $\mathfrak{p}_i$ is *isolated* (minimal) if it is minimal in $\{\mathfrak{p}_1, \ldots, \mathfrak{p}_n\}$, otherwise *embedded*. Let $S(N)$ denote the contraction in M of $S^{-1}N \subseteq S^{-1}M$ (i.e. $\{m : sm \in N \text{ for some } s \in S\}$), for a submodule N.

(4.6′) *Any prime ideal $\mathfrak{p} \supseteq \mathrm{Ann}(M)$ contains a minimal prime belonging to 0, so the minimal primes of $\mathrm{Ann}(M)$ (minimal among primes containing $\mathrm{Ann}\, M$) are exactly the minimal elements of $\{\mathfrak{p}_1, \ldots, \mathfrak{p}_n\}$.* Proof: by Ex. 20, $r(\mathrm{Ann}\, M) = r_M(0) = \bigcap \mathfrak{p}_i$. If $\mathfrak{p} \supseteq \mathrm{Ann}(M)$ then $\mathfrak{p} \supseteq r(\mathrm{Ann}\, M) = \bigcap \mathfrak{p}_i \supseteq \prod \mathfrak{p}_i$, so $\mathfrak{p} \supseteq \mathfrak{p}_i$ for some i (1.11). Also each $\mathfrak{p}_i \supseteq \bigcap \mathfrak{p}_j = r(\mathrm{Ann}\, M)$ contains $\mathrm{Ann}\, M$. Hence the minimal primes over $\mathrm{Ann}\, M$ are the minimal $\mathfrak{p}_i$.

(4.7′) *The set of zero-divisors in M is $\bigcup_{i=1}^{n} \mathfrak{p}_i = \{x \in A : (0 :_M x) \neq 0\}$, i.e. $\bigcup_{m \neq 0} r(0 : m)$.* Proof: if x is a zero-divisor, $xm = 0$ with $m \neq 0$, so $x \in (0 : m) \subseteq r(0 : m) = \bigcap_{m \notin Q_i} \mathfrak{p}_i \subseteq \mathfrak{p}_j$ for some j (there is at least one i with $m \notin Q_i$ since $m \neq 0$). Conversely, by Ex. 22, $\mathfrak{p}_i = r(0 : m_i)$ with $m_i \neq 0$ and $(0 : m_i)$ $\mathfrak{p}_i$-primary; if $x \in \mathfrak{p}_i$ then $x^k m_i = 0$ for some k; with k minimal, $x^{k-1} m_i \neq 0$ is killed by x, so x is a zero-divisor. (Applied to $M = A$ this is 4.7.)

(4.8′) *Let S be multiplicatively closed and Q a $\mathfrak{p}$-primary submodule of M. (i) If $S \cap \mathfrak{p} \neq \emptyset$ then $S^{-1}Q = S^{-1}M$. (ii) If $S \cap \mathfrak{p} = \emptyset$ then $S^{-1}Q$ is $S^{-1}\mathfrak{p}$-primary in $S^{-1}M$ and its contraction in M is Q.* Proof: (i) Let $s \in S \cap \mathfrak{p}$, $s^k M \subseteq Q$. Then $m/t = s^k m / s^k t \in S^{-1}Q$. (ii) *Contraction.* If $m/1 = q/s$ $(q \in Q)$, then $u(sm - q) = 0$ for some $u \in S$, so $usm \in Q$ and $us \notin \mathfrak{p}$, hence $m \in Q$ by (4.4′)(iii). So the contraction is Q; in particular $S^{-1}Q \neq S^{-1}M$. *Primary:* let a/s be a zero-divisor in $S^{-1}M/S^{-1}Q = S^{-1}(M/Q)$: $(a/s)(m/t) = 0$ with $m/t \neq 0$, i.e. $uam \in Q$ for some $u \in S$ while $um \notin Q$ (else $m/t = 0$). So a is a zero-divisor on M/Q, hence $a^k M \subseteq Q$, and $(a/s)^k$ kills $S^{-1}M/S^{-1}Q$. *Radical:* if $(x/s)^k S^{-1}M \subseteq S^{-1}Q$, then for each m, $x^k m/s^k = q/t$, so $u\,t\,x^k m \in Q$ for some $u \in S$; with $ut \notin \mathfrak{p}$ we get $utx^k \in (Q : M) \subseteq \mathfrak{p}$, so $x \in \mathfrak{p}$. Conversely $x \in \mathfrak{p} \Rightarrow x^k M \subseteq Q$. So $r_{S^{-1}M}(S^{-1}Q) = S^{-1}\mathfrak{p}$.

(4.9′) *Let $0 = \bigcap Q_i$ be minimal and S meet $\mathfrak{p}_{m+1}, \ldots, \mathfrak{p}_n$ but not $\mathfrak{p}_1, \ldots, \mathfrak{p}_m$. Then $S^{-1}0 = \bigcap_{i \leq m} S^{-1}Q_i$ and $S(0) = \bigcap_{i \leq m} Q_i$, and these are minimal primary decompositions.* Proof: $0 = S^{-1}0 = \bigcap_i S^{-1}Q_i$ (3.4) $= \bigcap_{i \leq m} S^{-1}Q_i$ by (4.8′)(i), and each $S^{-1}Q_i$ $(i \leq m)$ is $S^{-1}\mathfrak{p}_i$-primary by (4.8′)(ii); the $S^{-1}\mathfrak{p}_i$ are distinct since the $\mathfrak{p}_i$ are. Irredundant: if $\bigcap_{j \neq i, j \leq m} S^{-1}Q_j \subseteq S^{-1}Q_i$, contracting gives $\bigcap_{j \neq i, j \leq m} Q_j \subseteq Q_i$, hence $\bigcap_{j \neq i} Q_j \subseteq Q_i$, contradicting minimality of the original decomposition. Contracting the decomposition gives $S(0) = \bigcap_{i \leq m} Q_i$ by (4.8′)(ii).

(4.10′) *(Second uniqueness theorem.) Let $\{\mathfrak{p}_{i_1}, \ldots, \mathfrak{p}_{i_m}\}$ be an isolated set of primes belonging to 0 (a set Σ such that $\mathfrak{p}'$ belonging to 0, $\mathfrak{p}' \subseteq \mathfrak{p} \in \Sigma$ implies $\mathfrak{p}' \in \Sigma$). Then $Q_{i_1} \cap \cdots \cap Q_{i_m}$ is independent of the decomposition.* Proof: let $S = A - \bigcup_{\mathfrak{p} \in \Sigma} \mathfrak{p}$, multiplicatively closed. For $\mathfrak{p}'$ belonging to 0: $\mathfrak{p}' \in \Sigma \Rightarrow \mathfrak{p}' \cap S = \emptyset$; and $\mathfrak{p}' \notin \Sigma \Rightarrow \mathfrak{p}' \not\subseteq \bigcup_{\mathfrak{p} \in \Sigma} \mathfrak{p}$ (by 1.11 and isolatedness), so $\mathfrak{p}' \cap S \neq \emptyset$. By (4.9′), $Q_{i_1} \cap \cdots \cap Q_{i_m} = S(0)$, which depends only on 0 and Σ (the $\mathfrak{p}_i$ being intrinsic by Ex. 22).

(4.11′) *The isolated primary components (those Q_i with $\mathfrak{p}_i$ minimal) are uniquely determined by M.* Proof: the set of minimal $\mathfrak{p}_i$ is isolated, and for a single minimal $\mathfrak{p}_i$ the set $\{\mathfrak{p}_i\}$ is isolated; by (4.10′) $Q_i = S(0)$ with $S = A - \mathfrak{p}_i$ (which meets $\mathfrak{p}_j$ iff $\mathfrak{p}_j \not\subseteq \mathfrak{p}_i$), which is independent of the decomposition.

Remark. Taking $M = A$ and N an ideal recovers 4.6–4.11 (via $M = A/\mathfrak{a}$). $\square$